

NYC Airbnb Price Prediction

Undergraduate Empirical Project - ECN 190D, UC Davis

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Project Summary

This project uses 2019 New York City Airbnb listing data from Kaggle to predict listing prices based on room type, borough, neighborhood, minimum-night requirements, and number of reviews. The analysis combines exploratory data analysis with bootstrapping, cross-validation, and LASSO regression to evaluate pricing patterns and out-of-sample predictive performance.

Skills demonstrated: R, predictive modeling, LASSO regression, cross-validation, bootstrapping, dummy variables, data visualization, and applied economic analysis.

Introduction

New York City receives tens of millions of visitors each year, and Airbnb has become an important part of the city's hospitality market. The platform began as a way for individuals to rent extra space in their homes, but it has also created opportunities for property owners to operate multiple listings in ways that resemble hotel activity. This shift has generated both benefits and concerns: Airbnb can bring tourist spending into local neighborhoods, but it can also reduce the supply of housing available to residents and place upward pressure on rents.

This paper examines 2019 Airbnb listings in New York City and uses statistical learning techniques to predict listing prices. The analysis focuses on room type, borough, neighborhood, minimum nights, and number of reviews. Bootstrapping is used to estimate the sampling variability of regression coefficients, and cross-validated LASSO is used to evaluate model performance on out-of-sample data.

Literature Review and Context

Airbnb was founded in 2008 by Brian Chesky, Joe Gebbia, and Nathan Blecharczyk. The company grew from a small short-term rental idea into a major hospitality platform. However, the growth of Airbnb has raised concerns about its effects on housing markets. In New York City, some hosts operate multiple entire-home listings or large numbers of private rooms, creating a commercial short-term rental market within the broader housing system.

The New York City Comptroller's 2018 report argued that Airbnb can remove housing units from the long-term rental supply and contribute to higher rents. The report estimated that for each one percent of residential units in a neighborhood listed on Airbnb, neighborhood rental rates rose by 1.58 percent. This context motivates the present analysis of Airbnb listing prices and the factors that help explain price variation across New York City.

Data

The dataset contains 48,895 Airbnb observations from New York City in 2019. Although the raw data includes variables such as host name, host ID, and listing name, this analysis focuses on variables most directly related to price prediction: neighborhood, neighborhood group, room type, minimum-night stay, and number of reviews. Categorical variables such as neighborhood, neighborhood group, and room type were converted into factor variables before modeling.

The distribution of listings by borough is heavily concentrated in Manhattan and Brooklyn. The dataset includes 21,661 listings in Manhattan, 20,104 in Brooklyn, 5,666 in Queens, 1,091 in the Bronx, and 373 in Staten Island. Average nightly prices also vary substantially across boroughs: Manhattan has the highest average price at \$196.88 per night, followed by Brooklyn at \$123.38, Staten Island at \$114.81, Queens at \$99.51, and the Bronx at \$87.50.

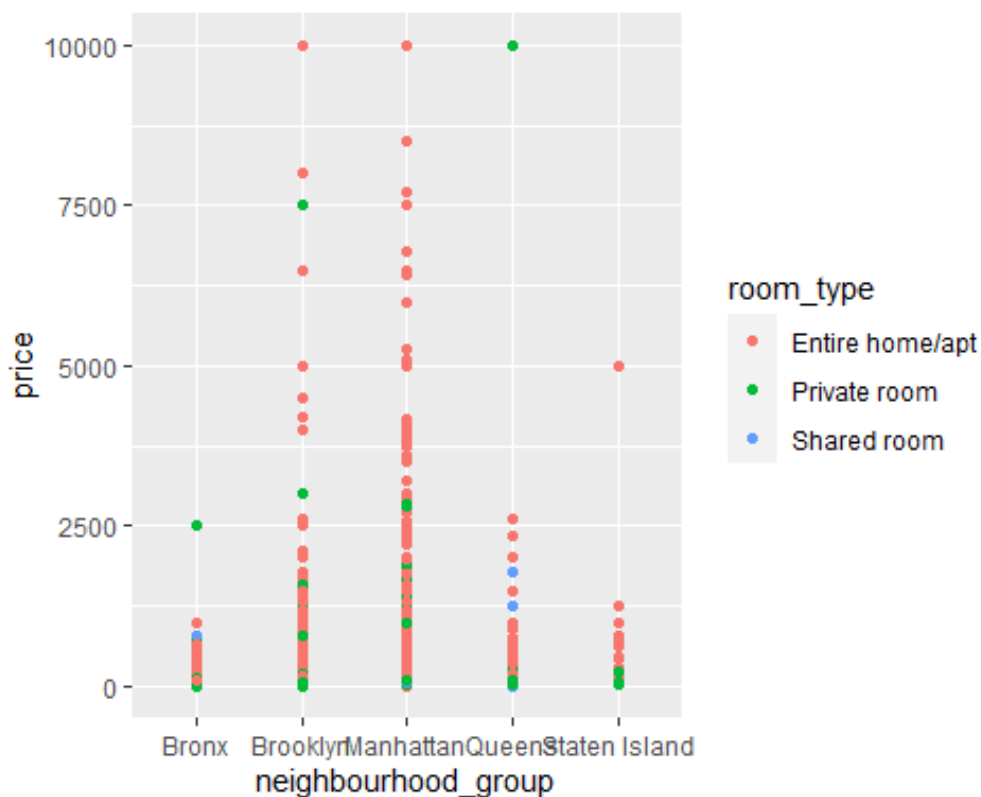


Figure 1. Distribution of Airbnb prices by borough.

The visualization shows that most listings are clustered below \$2,500 per night, with Manhattan containing the highest-cost observations.

Methodology

Bootstrap Standard Errors

Bootstrapping was used to estimate the sampling variability of the coefficients from a generalized linear model. The bootstrap repeatedly resamples from the observed data with replacement and re-estimates the model across many simulated samples. This produces a distribution of coefficient estimates that can be used

to approximate standard errors. To reduce computing time, the bootstrap was run on the first 10,000 observations using 500 repetitions.

Cross-Validation

Cross-validation was used to evaluate out-of-sample predictive performance. The model was trained using five folds: four folds were used to fit the model, and the remaining fold was used to estimate prediction error. This process was repeated so that each fold served as the validation set. The resulting errors provide an estimate of how well the model performs on data it has not directly observed.

LASSO Regression

LASSO, or Least Absolute Shrinkage and Selection Operator, was used as a regularized regression method. LASSO penalizes the absolute size of coefficients, shrinking some coefficients toward zero and helping control model complexity. Lower values of the penalty parameter allow a more complex model, while higher values create a simpler model. The cross-validated LASSO model selected the penalty value that minimized out-of-sample deviance.

Results

The generalized linear model included room type and neighborhood group dummy variables. The base room type was entire home/apartment, and the base borough was the Bronx. The table below summarizes selected coefficient estimates from the model.

Variable	Estimate	Std. Error
Private room	-98.381768	2.201461
Shared room	-120.789955	6.958869
Brooklyn	13.494093	39.521860
Manhattan	129.337054	36.061894
Queens	4.155408	38.108762
Staten Island	543.502495	230.141847

Because specific neighborhood indicators include more than 200 categories, the table reports selected higher-level coefficients. The estimates indicate that, relative to an entire home/apartment, private rooms and shared rooms are associated with lower prices. Manhattan listings are associated with higher prices relative to the Bronx, consistent with the descriptive statistics.

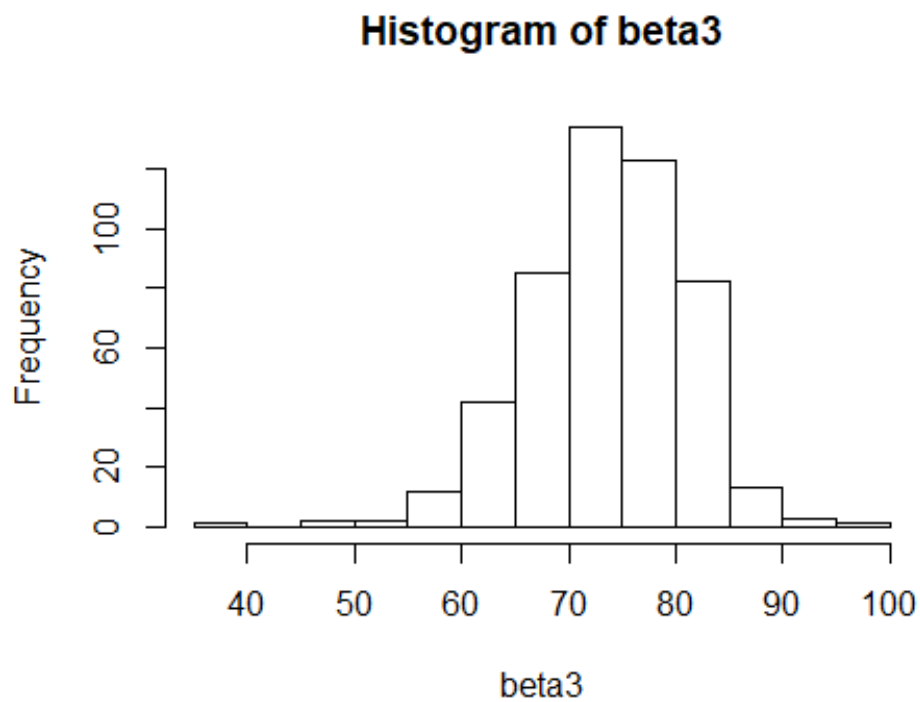


Figure 2. Bootstrap distribution of the Manhattan coefficient.

The bootstrap distribution provides a visual check on the variability of the Manhattan coefficient across repeated resamples.

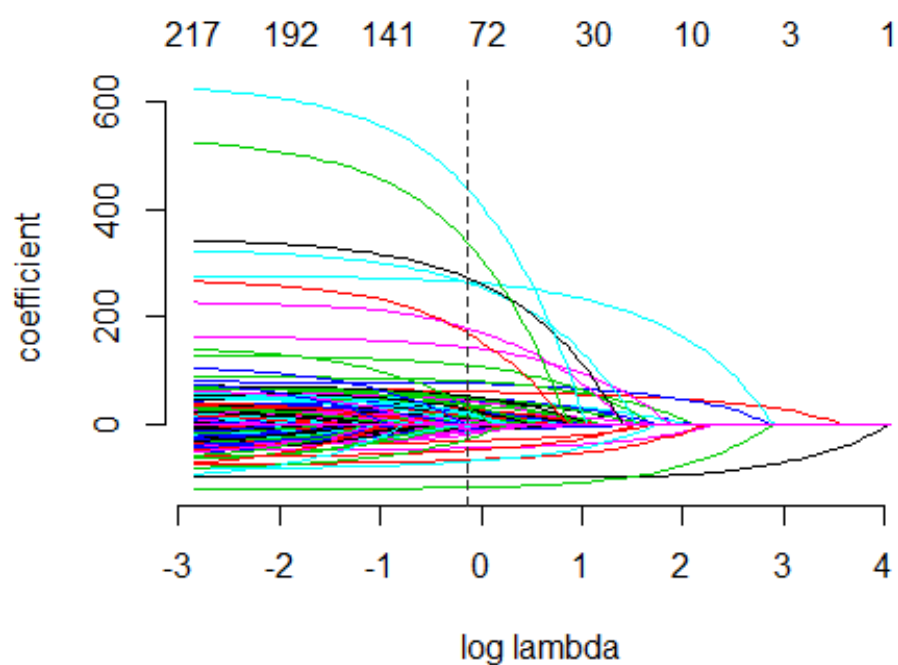


Figure 3. Cross-validated LASSO coefficient path.

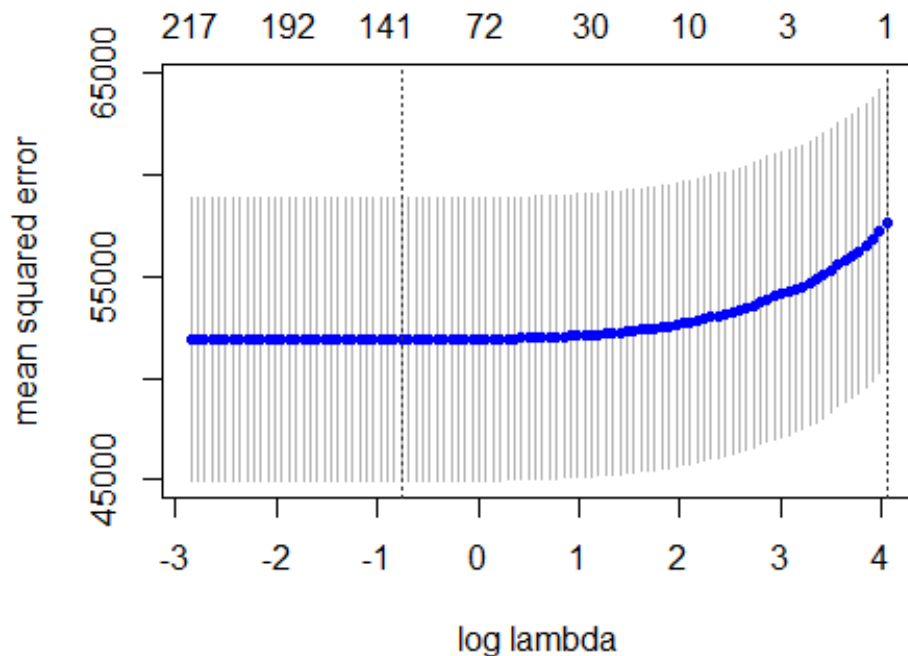


Figure 4. Mean out-of-sample error across penalty values.

The cross-validated LASSO selected a model with a minimum out-of-sample deviance at segment 68. The resulting out-of-sample R-squared was approximately 0.538, meaning that the model explained slightly more than half of the variation in out-of-sample Airbnb prices. Location appears to explain a large share of price variation, although other omitted factors such as amenities, views, proximity to public transit, and listing quality likely explain additional variation.

Conclusion

This project used cross-validation and LASSO regression to predict Airbnb prices in New York City. The results suggest that location and room type are important predictors of listing price, with Manhattan and entire-home listings associated with higher prices. The model's out-of-sample R-squared of approximately 0.538 indicates meaningful predictive power, while also leaving substantial unexplained variation. Factors not included in the dataset, such as amenities, views, transit access, building quality, and host reputation, likely contribute to the remaining variation in price.

The project demonstrates how statistical learning tools can be applied to a practical business and housing-market question. It also highlights the value of out-of-sample evaluation when building predictive models.

Works Cited

"Airbnb Statistics [2020]: User & Market Growth Data." IPropertyManagement.com, Mar. 2020, ipropertymanagement.com/research/airbnb-statistics.

Boone, Alastair. "What Airbnb Did to New York City." CityLab, 25 June 2018, www.citylab.com/equity/2018/03/what-airbnb-did-to-new-york-city/552749/.

Eren, Selçuk. "The Impact of Airbnb on NYC Rents." Comptroller.nyc.gov, 3 May 2018, comptroller.nyc.gov/reports/the-impact-of-airbnb-on-nyc-rents/.

González-Rivera, Christian. "Destination New York." Center for an Urban Future, May 2018, nycfuture.org/research/destination-new-york.

"History of Airbnb." Airbnb, 16 Dec. 2016, airbnb370.wordpress.com/history-of-airbnb/.

Taddy, Matt. Business Data Science: Combining Machine Learning and Economics to Optimize, Automate, and Accelerate Business Decisions.